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Quantitative Ester Analysis in Cachaça and Distilled Spirits by Gas Chromatography−Mass Spectrometry (GC−MS)

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An analytical procedure for the separation and quantification of ethyl acetate, ethyl butyrate, ethyl hexanoate, ethyl lactate, ethyl octanoate, ethyl nonanoate, ethyl decanoate, isoamyl octanoate, and ethyl laurate in cachaça, rum, and whisky by direct injection gas chromatography−mass spectrometry was developed. The analytical method is simple, selective, and appropriated for the determination of esters in distilled spirits. The limit of detection ranged from 29 (ethyl hexanoate) to 530 (ethyl acetate) μg L^{-1}, whereas the standard deviation for repeatability was between 0.774% (ethyl hexanoate) and 5.05% (isoamyl octanoate). Relative standard deviation values for accuracy vary from 90.3 to 98.5% for ethyl butyrate and ethyl acetate, respectively. Ethyl acetate was shown to be the major ester in cachaça (median content of 22.6 mg 100 mL^{-1} anhydrous alcohol), followed by ethyl lactate (median content of 8.32 mg 100 mL^{-1} anhydrous alcohol). Cachaça produced in copper and hybrid alembic present a higher content of ethyl acetate and ethyl lactate than those produced in a stainless-steel column, whereas cachaça produced by distillation in a stainless-steel column present a higher content of ethyl octanoate, ethyl decanoate, and ethyl laurate. As expected, ethyl acetate is the major ester in whiskey and rum, followed by ethyl lactate for samples of rum. Nevertheless, whiskey samples exhibit ethyl lactate at contents lower or at the same order of magnitude of the fatty esters.

KEYWORDS: Cachaça; GC−MS; ester; spirits; ethyl acetate; ethyl lactate

INTRODUCTION

The Brazilian sugar-cane spirit, cachaça, is produced from the distillation of the fermented sugar-cane juice and has an ethanol content between 38 and 48% (v/v) (*1*). The annual production of cachaça is around 2 billion liters, of which less than 1% is exported (*2*). Recent efforts have been made to improve the quality of cachaça in view to expand its dissemination in the international trade market. As part of our research efforts, a better knowledge of the chemical composition of cachaça has been gained (*2–9*). Although on some extension the qualitative composition of cachaça has been previously subject of discussion, as far as we know, the quantitative profile of the fatty esters in cachaça has not yet been reported.

Similar to other distilled spirits, cachaça is organoleptically characterized by the content of secondary metabolites, also known as congeners, such as higher alcohols, esters, carboxylic acids, and carbonyl compounds, which contribute to the peculiar flavor of the spirit (*9–11*). Esters are well-known aroma compounds occurring at relatively high content and exhibiting a low odor threshold and therefore are quite relevant for the beverage sensory properties (*9–13*).

With the aim of improving the chemical characterization of cachaça, this paper reports a selective and sensitive method for the quantitative determination of eight ethyl esters (ethyl acetate, ethyl butyrate, ethyl hexanoate, ethyl lactate, ethyl octanoate, ethyl nonanoate, ethyl decanoate, and ethyl laurate) and isoamyl octanoate in distilled spirits by direct injection gas chromatography−mass spectrometry (GC−MS) analysis. These esters are known to be the most abundant esters in spirits, and their presence is commonly related to the pleasant fruity bouquet of alcoholic beverages (*14–16*).

EXPERIMENTAL SECTION

Materials. Ester standards (ethyl acetate, ethyl butyrate, ethyl hexanoate, ethyl lactate, ethyl octanoate, ethyl nonanoate, ethyl decanoate, isoamyl octanoate, and ethyl laurate) and the internal standard 4-methyl-2-pentanol were purchased from Sigma-Aldrich (Steinheim, Germany) and used as received. The high-performance liquid chromatography (HPLC)-grade ethanol was obtained from Fischer Scientific (Fair Lawn, NJ) and used without further purification. Water was previously bidistilled and further deionized using a Milli-Q system Millipore (Millipore, Bedford, MA).

GC−MS Analysis. Sample analyses were carried out in a gas chromatograph model GC17A (Shimadzu, Tokyo, Japan) hyphened to a mass selective detector model QP5050A (Shimadzu, Tokyo, Japan) using electron impact (70 eV) as the ionization source. The target analytes were separated through a capillary column coated with an

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